

Skill Assessment

SQL Table

cars

No	Field Name
1	id
2	name
3	manufacture_id
4	date_created

manufactures

No	Field Name
1	id
2	name

By using PHP version 7 or 8 and MYSQL / MariaDB (you can use any framework), please do SQL query for the SQL table above. The final result must be look like this:

id	Model name	date created	manufacture
1	S70	21 Jan 2024 1:19 PM	Proton
2	Civic	11 Feb 2024 1:19 PM	Honda
3	Altis	21 Feb 2024 9:19 AM	Toyota

Publish the completed source code on github.com and share the repository url to azrul@mytsg.com.my cc rozana@mytsg.com.my

Answer:

1. Database Setup

- Create the two tables:

```
CREATE TABLE manufactures (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(255)  
);
```

```
CREATE TABLE cars (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(255),  
  manufacture_id INT,  
  date_created DATETIME,  
  FOREIGN KEY (manufacture_id) REFERENCES manufactures(id)  
);
```

- Insert sample data into table cars and manufactures:

```
INSERT INTO manufactures (name) VALUES  
('Proton'), ('Honda'), ('Toyota');
```

```
INSERT INTO cars (name, manufacture_id, date_created) VALUES  
('S70', 1, '2024-01-21 13:19:00'),  
('Civic', 2, '2024-02-11 13:19:00'),  
('Altis', 3, '2024-02-21 09:19:00');
```

2. SQL Query

- Write the query to join and format data:

```
SELECT  
  cars.id,  
  cars.name AS 'Model name',  
  DATE_FORMAT(cars.date_created, '%d %b %Y %l:%i %p') AS 'date created',  
  manufactures.name AS 'manufacture'  
FROM cars  
INNER JOIN manufactures ON cars.manufacture_id = manufactures.id;
```

3. PHP Script

- Connect to the database and execute the query:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">
```

```

<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Car Details</title>
<!-- Add Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha3/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
  <div class="container mt-5">
    <h1 class="text-center mb-4">Car Details</h1>
    <div class="card">
      <div class="card-body">
        <table class="table table-bordered table-striped">
          <thead class="table-dark">
            <tr>
              <th>ID</th>
              <th>Model Name</th>
              <th>Date Created</th>
              <th>Manufacture</th>
            </tr>
          </thead>
          <tbody>
            @foreach ($cars as $car)
              <tr>
                <td>{{ $car->id }}</td>
                <td>{{ $car->'Model name' }}</td>
                <td>{{ $car->'Date created' }}</td>
                <td>{{ $car->Manufacture }}</td>
              </tr>
            @endforeach
          </tbody>
        </table>
        @if (count($cars) === 0)
          <p class="text-center text-muted">No cars found.</p>
        @endif
      </div>
    </div>
  </div>
</body>
</html>

```

Result:

1. Cars table

✓ Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

`SELECT * FROM `cars``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	name	manufacture_id	date_created
☐	Edit	Copy	Delete	1	S70	1	2024-01-21 13:19:00
☐	Edit	Copy	Delete	2	Civic	2	2024-02-11 13:19:00
☐	Edit	Copy	Delete	3	Altis	3	2024-02-21 09:19:00

↑ ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

2. SQL Query

Query to create the two table cars and manufactures

Run SQL query/queries on table tsg.cars: ⓘ

```
1 CREATE TABLE manufactures (  
2   id INT AUTO_INCREMENT PRIMARY KEY,  
3   name VARCHAR(255)  
4 );  
5  
6 CREATE TABLE cars (  
7   id INT AUTO_INCREMENT PRIMARY KEY,  
8   name VARCHAR(255),  
9   manufacture_id INT,  
10  date_created DATETIME,  
11  FOREIGN KEY (manufacture_id) REFERENCES manufactures(id)  
12 );  
13
```

SELECT * | SELECT | INSERT | UPDATE | DELETE | Clear | Format | Get auto-saved query

SQL Query to insert sample data given

Run SQL query/queries on table `tsg.cars`: ⓘ

```
1 INSERT INTO manufactures (name) VALUES
2 ('Proton'), ('Honda'), ('Toyota');
3
4 INSERT INTO cars (name, manufacture_id, date_created) VALUES
5 ('S70', 1, '2024-01-21 13:19:00'),
6 ('Civic', 2, '2024-02-11 13:19:00'),
7 ('Altis', 3, '2024-02-21 09:19:00');
8 |
```

SELECT *

SELECT

INSERT

UPDATE

DELETE

Clear

Format

Get auto-saved query

SQL Query to join and format data

Run SQL query/queries on table `tsg.cars`: ⓘ

```
1 SELECT
2     cars.id,
3     cars.name AS 'Model name',
4     DATE_FORMAT(cars.date_created, '%d %b %Y %l:%i %p') AS 'date created',
5     manufactures.name AS 'manufacture'
6 FROM cars
7 INNER JOIN manufactures ON cars.manufacture_id = manufactures.id;
8 |
```

SELECT *

SELECT

INSERT

UPDATE

DELETE

Clear

Format

Get auto-saved query

3. Manufactures table

Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

```
SELECT * FROM `manufactures`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

					id	name
☐	Edit	Copy	Delete		1	Proton
☐	Edit	Copy	Delete		2	Honda
☐	Edit	Copy	Delete		3	Toyota

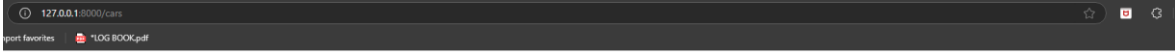
[Check all](#) | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

4. PHP Script

```
resources > views > cars > index.blade.php > html > body > div.container.mt-5 > div.card > div.card-body > table.table.table-bordered.table-striped >
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
6 <title>Car Details</title>
7 <!-- Add Bootstrap CSS -->
8 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha3/dist/css/bootstrap.min.css" rel="stylesheet">
9 </head>
10 <body>
11 <div class="container mt-5">
12 <h1 class="text-center mb-4">Car Details</h1>
13 <div class="card">
14 <div class="card-body">
15 <table class="table table-bordered table-striped">
16 <thead class="table-dark">
17 <tr>
18 <th>ID</th>
19 <th>Model Name</th>
20 <th>Date Created</th>
21 <th>Manufacture</th>
22 </tr>
23 </thead>
24 <tbody>
25 @foreach ($cars as $car)
26 <tr>
27 <td>{{ $car->id }}</td>
28 <td>{{ $car->{'Model name'} }}</td>
29 <td>{{ $car->{'Date created'} }}</td>
30 <td>{{ $car->Manufacture }}</td>
31 </tr>
32 @endforeach
33 </tbody>
34 </table>
```

5. Result



Car Details

ID	Model Name	Date Created	Manufacture
1	S70	2024-01-21 13:19:00	Proton
2	Civic	2024-02-11 13:19:00	Honda
3	Altis	2024-02-21 09:19:00	Toyota